Zeno: Product and Technical Plan

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Product description

Zeno is a secure video and live-class platform for Indian educators. It lets teachers and coaching institutes upload lectures, run live classes and sell courses, while keeping their content protected from piracy and smooth to watch on weak mobile networks. It enters the market through solo teachers and small institutes, and is designed to grow, stage by stage, to the protection, scale and reliability of established platforms such as TPStreams.

What is planned

- Protected video hosting: encrypted streaming, per-viewer watermarking, signed links and session limits, growing to Widevine and FairPlay DRM and forensic watermarking.
- Live classes from OBS that turn into course lessons automatically when the class ends.
- Built-in course selling: course pages, UPI and card checkout, and automatic enrollment.
- Playback built for Indian networks: low-start adaptive streaming and an audio-only mode.
- Creator analytics, then native Android and iOS SDKs with offline viewing.

How it is planned

- A Rust control plane (API, licence proxy, billing) kept separate from an FFmpeg-based media plane (transcoding, packaging, live), so each scales on its own.
- Encrypted HLS/CMAF stored in object storage and delivered through a CDN; application servers handle only manifests, keys and licences.
- Four stages over about two years: an MVP in roughly three months, then protection parity (multi-DRM, mobile SDKs), then scale (Kubernetes, multi-CDN, forensic watermarking), then enterprise readiness (direct DRM licences, compliance, 99.99% availability).
- Every MVP component sits behind an interface that the target component can replace without a rewrite.

What is taken into account

- **Security:** token-gated keys, signed URLs, verified payment webhooks, role checks, and an automated negative test suite in CI.
- **Content protection limits:** the MVP does not claim DRM parity; the path to real DRM is planned and staged.

- **Cost:** bandwidth, transcoding, DRM licences and people are the main cost drivers, priced from provider quotes before scaling.
- **Reliability:** redundant live ingest, health checks, autoscaling and on-call, with availability targets per stage.
- **Compliance:** India's DPDP Act from launch; SOC 2 or ISO 27001 before enterprise sales.
- **Licensing:** DRM via a commercial provider first, since direct Widevine and FairPlay approval takes months and is not guaranteed.
- **Quality:** tests written before code, line-by-line review of security-critical modules, and real-device testing on low-end Android, iOS and desktop.

What success looks like for Stage 1

A flawless end-to-end demo (upload, pay, watch with watermark, live class saved as a lesson); measured numbers for upload-to-ready time, playback start on slow 4G, cost per student-hour and concurrent viewers; and 5–10 pilot creators using the platform, with some paying.

Product overview

Zeno is a secure video platform for Indian educators, built to take the solo teachers and small institutes that TPStreams prices and designs past. Creators upload lectures, run live classes and sell courses, with content that stays protected and plays well on weak mobile networks. *Zeno* is a working name.

Positioning. TPStreams-style protected video for the solo teacher: secure playback, live classes and course sales in one product, from a much lower entry price and with no technical team needed.

Main competitor: TPStreams. TPStreams is the established Indian platform for secure education video, and it is strong on content protection and scale. Zeno does not try to out-DRM it at launch; it aims to win the segment that TPStreams' pricing and focus leave open. TPStreams figures below are the company's own claims from its website, checked on 1 November 2026.

Dimension	TPStreams	Zeno (plan)
Entry price	$150 a month (Start plan)	Lower entry tier for solo teachers, set after measuring cost to serve
Built for	Large exam-prep, certification and regulatory programmes	Solo teachers and small coaching institutes

Dimension	TPStreams	Zeno (plan)
Content protection	Widevine and FairPlay DRM, forensic watermarking, token auth and domain lock on every plan	Widevine and FairPlay multi-DRM (Stage 2) and forensic watermarking (Stage 3), on top of AES-128 and a visible watermark from the MVP
Live streaming	Included; 24 live hours on the Start plan	Included; the recording becomes a course lesson automatically
Course sales	Video platform; LMS offered as a separate Testpress product	Course pages and checkout built in
Weak networks	Adaptive bitrate tuned for 4G and unstable networks	Low-start playback plus an audio-only mode
Support	Email on Start; WhatsApp from Pro ($450 a month)	WhatsApp support from the first tier

Gaps Zeno targets

- **Price of entry:** $150 a month is a heavy fixed cost for a teacher with one batch of students, and the plan is sized for far more traffic than they use.
- **Video and selling in one place:** a solo teacher wants to upload, price and sell without stitching a video platform to a separate LMS.
- **Teacher workflow:** a live class should become a finished lesson with no manual export or re-upload.
- **Self-serve start:** sign up and publish the first course the same day, without a sales demo.

How Zeno wins. Three focused bets, rather than matching TPStreams feature for feature:

1. **Cheaper and simpler to start:** a low entry tier, self-serve signup, and course sales built in, so a single teacher can go live in a day.
2. **Traceable playback:** every viewer sees a moving watermark with their name and phone number, so a leaked recording points back to its source.
3. **Built for the teacher's day:** live class becomes a lesson automatically, and students on patchy 4G get low-start playback and audio-only mode.

Path to TPStreams' level. Zeno enters through the solo-teacher segment, then closes the gap in stages: multi-DRM and mobile SDKs in Stage 2, multi-CDN and forensic watermarking in Stage 3, direct DRM licences, compliance certification and 99.99% availability in Stage 4 (see the delivery roadmap). Until Stage 2 ships, Zeno will not claim

protection parity. The main competitive risk is TPStreams launching a cheaper creator tier; Zeno's defence is built-in selling and a teacher workflow that a video-infrastructure product does not offer.

Target users

The first customer is the solo teacher or small coaching institute selling exam-prep or skill courses; students and live-class teachers are the users they bring with them.

Persona	Who they are	What they need from Zeno	What makes them leave
Creator (paying customer)	Solo teacher or institute with 1–5 staff, 50–2,000 students	Upload once, sell access, stop leaks, simple monthly bill	Leaked videos, surprise bandwidth bills, complex setup
Student (end user)	Learner on a mid-range Android phone, often on mobile data	Videos that start quickly and do not buffer, resume where they stopped	Buffering, login friction, data usage
Live teacher	The creator or a hired faculty member running scheduled classes	Go live from OBS with one stream key, recording saved automatically	Failed streams, manual re-uploads after class

The student-count range above is a planning assumption, to be confirmed in creator interviews before the build starts.

Features

The MVP covers the full loop of upload, sell, watch and teach live; everything else waits until paying creators ask for it.

Feature	For	Release
Account sign-up, login, roles (creator, student, admin)	All	MVP
Video upload with progress and resumable transfer	Creator	MVP

Feature	For	Release
Automatic transcoding to 144p–720p plus audio-only	Creator, student	MVP
Courses, ordered lessons, pricing	Creator	MVP
Checkout with UPI and cards, automatic enrollment	Student	MVP
Encrypted playback with per-user, short-lived access	Student	MVP
Dynamic watermark with viewer name and phone	Creator	MVP
Low-bandwidth mode and resume-from-last-position	Student	MVP
Live class from OBS with per-class stream key	Live teacher	MVP
Recording automatically added as a lesson	Live teacher	MVP
Basic analytics: views, watch time, drop-off per lesson	Creator	MVP
Concurrent-session limit per student	Creator	MVP
Live chat during class	Student	Next
Auto-generated captions and chapters	Student	Next
Installable mobile app (PWA) with offline viewing	Student	Next
Quizzes and notes attached to lessons	Creator	Later
Custom domain and branding per creator	Creator	Later
Multi-DRM (Widevine, FairPlay)	Creator	Stage 2

How it works for users

Each persona has one main journey, and all three end with a student watching a protected lesson.

Creator publishes a course

1. Signs up, creates a course with a title, description and price.
2. Uploads lecture files; each shows processing status until it is ready to publish.
3. Orders lessons, previews them as a student would, and publishes the course.

4. Shares the course link on WhatsApp, Instagram or YouTube.

Student buys and watches

1. Opens the course link, signs up with phone or email.
2. Pays through UPI or card; enrollment is created as soon as the payment is confirmed.
3. Opens a lesson; playback starts at a low quality and steps up as the network allows, with the watermark visible.
4. Leaves and returns later; playback resumes from the last position.

Teacher runs a live class

1. Schedules a class inside a course; Zeno issues a stream key and server URL.
2. Pastes them into OBS and starts streaming; enrolled students see the class go live.
3. Ends the stream; the recording is processed and appears as a new lesson in the same course.

System architecture

Zeno's target architecture matches what a TPStreams-level platform needs: multi-DRM, multi-CDN delivery, native mobile SDKs and an autoscaling media plane, split into a Rust control plane and a separate media plane. The MVP is a small subset of this design, built so each part can be swapped up without a rewrite.

Multi-DRM and multi-CDN are what lift Zeno to TPStreams' level



target architecture · 5 layers, 20 components

How a request flows. A player asks the API gateway for a lesson, receives a manifest with tokenised URLs, fetches CMAF segments through the multi-CDN (where edge workers check the token and pick the viewer's watermark variant), and requests a licence from the DRM proxy, which checks enrollment and session policy before forwarding to Widevine or FairPlay. Uploads and live streams go through the media plane, which encrypts with keys from the key management service and writes to object storage behind the origin shield.

Control plane versus media plane. The control plane (accounts, billing, licences, API) is low-traffic and must be consistent; the media plane (transcoding, packaging, delivery) is high-volume and scales horizontally. Keeping them separate lets the transcode farm scale on GPUs during upload peaks without touching the API.

What the MVP uses from this. One API service, one VOD worker, one live server, a single CDN, AES-128 instead of DRM, and PostgreSQL, Redis and object storage on two servers. Every MVP interface (key delivery, manifest URLs, job queue) is designed so the target component can replace it later.

Tech stack

The control plane is Rust and the dashboards are TypeScript; the media plane uses industry-standard encoding, packaging and DRM tools. Each row shows what the MVP runs and what the target platform runs.

Layer	MVP	Target platform	Why
Control plane	Rust, axum on tokio	Same, split into API, licence proxy and billing services	Memory-safe, fast, predictable under load
Orchestration	Docker Compose on two servers	Kubernetes with autoscaling, Terraform for all infrastructure	Scale transcoding and live capacity on demand
Transcoding	FFmpeg on CPU	FFmpeg with GPU encoding (NVENC), per-title ladders; HEVC and AV1 later	Lower cost per hour, better quality per bit
Packaging	FFmpeg HLS	Shaka Packager, CMAF segments for HLS and DASH	One set of segments serves every device
Encryption	AES-128 HLS	CENC (cbcs) with multi-DRM	Required for Widevine and FairPlay
DRM	None	Widevine and FairPlay via a multi-DRM provider, then direct licences	Hardware-backed protection, TPStreams parity
Keys	Database with restricted access	HashiCorp Vault or cloud KMS, HSM-backed	Keys never stored in plain form

Layer	MVP	Target platform	Why
Live	RTMP via MediaMTX, HLS	SRT and RTMP ingest with redundant inputs, LL-HLS, DVR	Lower latency, no single point of failure
Storage	Single bucket (S3-compatible)	Multi-region, versioned object storage	Durability and regional delivery
Delivery	One CDN	Multi-CDN with traffic steering, origin shield, edge workers	Uptime, price leverage, edge token checks
Database	PostgreSQL	PostgreSQL primary with read replicas, point-in-time recovery	Reliability and read scale
Queue and events	Redis	NATS or Redis Streams for jobs; Kafka or Redpanda for events	Separate job dispatch from analytics firehose
Analytics	PostgreSQL queries	ClickHouse fed by player beacons	Fast QoE and engagement queries at volume
Players	hls.js web player	Shaka Player (web), Media3 (Android), AVPlayer (iOS) SDKs	DRM and offline support on every platform
Dashboards	Next.js, TypeScript, Tailwind	Same	Fast UI development
AI	None	Whisper on GPU for captions and transcripts	Accessibility and search
Payments	Razorpay	Razorpay and Stripe, usage-based invoicing	India and international customers
Observability	Sentry, basic metrics	Prometheus, Grafana, Loki, OpenTelemetry, paging	SLOs and on-call

Video processing and delivery

All video, recorded or live, ends up as encrypted multi-bitrate HLS in object storage, served through a CDN; the API never streams video bytes itself.

Recorded lessons (VOD)

1. The browser uploads the file straight to object storage using a presigned multipart URL, so large files never pass through the API server.
2. The API records the video and pushes a transcode job onto the queue.
3. A worker inspects the file with ffprobe, then FFmpeg produces renditions at 144p, 240p, 360p, 480p and 720p (never above the source) plus an audio-only track.
4. Output is segmented into HLS with segments of a few seconds, and each video is encrypted with its own AES-128 key.
5. The video is marked ready; the key is stored server-side, never alongside the segments.

Live classes

1. OBS pushes RTMP to the ingest server; a Rust service checks the stream key against the scheduled class.
2. FFmpeg converts the stream into a rolling HLS playlist with short segments, giving lecture-grade latency of several seconds.
3. When the stream ends, the segments are finalised, passed through the same packaging and encryption steps as VOD, and attached to the course as a lesson.

Playback

- The player asks the API for the lesson; the API checks enrollment and returns a playlist whose segment and key URLs are signed and expire quickly.
- The player starts at a low rendition and adapts to the network; low-bandwidth mode caps quality and offers audio-only.
- Segments come from the CDN; only the small playlist and key requests reach Zeno's servers.

Operating rules

- Live and VOD transcoding run on separate workers, so an upload never slows a class in progress.
- Jobs are idempotent and retried with backoff; failures are recorded with the error.
- FFmpeg is always called with argument lists, never a shell string, and file paths are generated by the server.

Security and content protection

Zeno's target protection matches the features TPStreams publishes: hardware-backed multi-DRM, forensic watermarking, token authentication and domain locking on every plan. The MVP ships the encryption, token and visible-watermark layers first; multi-DRM

arrives in Stage 2 and forensic watermarking in Stage 3.

Layer	Mechanism	Stage
Encryption (pre-DRM)	AES-128 HLS, one key per video, keys released only for valid tokens	MVP
Visible watermark	Moving overlay with viewer name, phone and session id	MVP
Signed URLs and domain lock	Expiring playlist and segment URLs; player restricted to the creator's domains	MVP
Session limits	Cap on simultaneous playback sessions per account	MVP
Multi-DRM	Widevine (Android, Chrome, smart TVs) and FairPlay (Safari, iOS) over CENC cbcs; PlayReady if customers need it	Stage 2
Licence policy	Per-session licences bound to user, device and expiry; security-level and HDCP rules per plan	Stage 2
Key management	Content keys generated and held in KMS or HSM, rotated, never stored in plain form	Stage 2
Offline playback	Persistent licences for encrypted downloads in the Android and iOS SDKs	Stage 3
Edge token auth	Tokens verified at the CDN edge before any segment is served	Stage 3
Forensic watermark	A/B segment variants; the edge serves each viewer a unique sequence that identifies the account in a leaked copy	Stage 3

Platform security

- Passwords hashed with argon2; login and OTP endpoints rate limited by IP and account.
- Role checks on every route: a student can never reach another student's data or a creator dashboard.
- Payment webhooks verified by signature and processed idempotently; enrollment is granted only by this path.
- Secrets and video keys held outside the codebase and outside object storage.
- Dependencies audited in CI; known-vulnerable crates block the build.

Getting DRM licences. There are two routes, and Zeno takes them in order. First, a commercial multi-DRM licence provider, which is fast to integrate and billed per licence issued. Later, direct licensing: Widevine through a licence agreement with Google, and FairPlay by requesting FairPlay Streaming credentials from Apple. TPStreams states that it is a direct Widevine partner. Direct licensing takes months and approval is not guaranteed, so the licence proxy is built to switch providers without changing players or content.

Compliance. India's Digital Personal Data Protection Act applies from launch (consent, data minimisation, breach reporting). SOC 2 Type II or ISO 27001 is targeted before selling to large institutes, since enterprise buyers ask for it during procurement.

Data model and API

Eight tables carry the whole product; enrollment is the single source of truth for who can watch what.

Entity	Main fields	Relationships
User	id, name, phone, email, password_hash, role	Owns courses (creator) or enrollments (student)
Course	id, creator_id, title, description, price, status	Has many lessons
Lesson	id, course_id, position, video_id or live_session_id	Belongs to a course
Video	id, status, duration, renditions, key_ref	Referenced by lessons
LiveSession	id, course_id, stream_key_hash, scheduled_at, state	Produces a video when it ends
Enrollment	id, user_id, course_id, payment_id, expires_at	Grants access
Payment	id, provider_ref, amount, status, raw_event	Creates an enrollment
PlaybackSession	id, user_id, video_id, device, ip, started_at, last_position	Watermark tracing, session limits, resume, analytics

REST API (v1)

Area	Endpoints
Auth	`POST /v1/auth/register`, `POST /v1/auth/login`, `POST /v1/auth/refresh`
Courses	`POST /v1/courses`, `GET /v1/courses/{id}`, `PATCH /v1/courses/{id}`, `POST /v1/courses/{id}/lessons`
Videos	`POST /v1/videos` (returns presigned upload), `GET /v1/videos/{id}` (status)
Playback	`GET /v1/play/{lesson_id}/master.m3u8`, `GET /v1/keys/{video_id}`, `POST /v1/play/{session_id}/heartbeat`
Commerce	`POST /v1/checkout/{course_id}`, `POST /v1/webhooks/razorpay`
Live	`POST /v1/live`, `GET /v1/live/{id}`, `GET /v1/live/{id}/index.m3u8`
Analytics	`GET /v1/analytics/courses/{id}`

Engineering approach

The code lives in one repository as a Rust workspace plus a web app, and nothing merges without passing tests, lint and a dependency audit.

Repository layout

```
zeno/
  crates/
    api/          axum HTTP server: auth, courses, playback, webhooks
    worker/       transcode job runner (FFmpeg)
    live/         stream-key check and live HLS control
    keyserver/    token validation and key delivery
    core/         shared types, database access, config
  web/            Next.js creator dashboard and student app
  deploy/         Docker Compose, Caddy, migrations
  tests/e2e/      Playwright end-to-end suites
```

Testing

Level	Tool	What it proves
Unit	cargo test	Token rules, playlist rewriting, enrollment logic
Integration	testcontainers (Postgres, Redis) and real FFmpeg on a short sample	Upload-to-ready pipeline, webhooks, key delivery
Security	Scripted negative tests	Expired, forged or cross-user tokens are rejected; webhooks cannot be spoofed
End to end	Playwright	Sign up, pay, watch with watermark, resume
Load	k6	Concurrent viewers and key requests before launch
Devices	Low-end Android on 4G, iPhone Safari, desktop Chrome	Playback and adaptive quality on real hardware

Quality gates

- Every pull request runs `cargo test`, `cargo clippy` and `cargo audit` in CI.
- Security-sensitive modules (auth, key server, webhooks, FFmpeg calls) require a line-by-line review.
- Database changes go through versioned migrations only.
- AI coding assistants are used for speed, with tests written first and generated code reviewed before merge.

Deployment, operations and cost

The MVP runs on two servers behind a CDN for an estimated ₹3,000–7,000 a month; the target platform runs on Kubernetes across regions, and at that stage people, not servers, are the largest cost. All figures are approximate and need confirming with providers.

Environments. Local (Docker Compose), staging (small VPS, test payments) and production. Every merge to main deploys to staging automatically; production deploys are manual, from tagged releases.

Production layout

- **App server:** API, key server, VOD worker, PostgreSQL and Redis in Docker Compose, with Caddy terminating HTTPS.
- **Live server:** RTMP ingest and live transcoding on a separate CPU-dedicated machine.

- **Storage and delivery:** encrypted segments in object storage, served through the CDN.
- **Backups:** nightly database dumps to object storage, with a restore tested monthly.

Monitoring. Error tracking, metrics on queue depth, transcode time, key-request rate and live-stream health, and an alert when a live stream drops or the job queue backs up.

Cost driver	MVP	Target platform
Compute	Two VPS, about ₹3,000–6,000 a month	Kubernetes cluster with an autoscaling CPU and GPU transcode pool
Delivery	Single CDN, small volume	Multi-CDN on committed volume; usually the largest infrastructure line
Storage	One bucket, a few hundred GB	Multi-region storage, grows with every customer library
DRM	None	Per-licence fees from a multi-DRM provider until direct licensing
Compliance	DPDP basics	SOC 2 or ISO 27001 audit and tooling
People	Solo founder	4–10 engineers across backend, media, mobile and SRE, plus support

Before raising or spending. Get written quotes from two multi-DRM providers, two CDNs and one GPU cloud for a modelled volume (for example 50 TB delivery and 500,000 licences a month), and build the plan's pricing on those numbers rather than estimates.

Reliability targets. 99.9% monthly availability in Stage 3, then 99.99% with multi-region failover in Stage 4. Live has redundant ingest and transcoding, and every service has health checks, autoscaling and alerting tied to an on-call rota.

Business model

Zeno charges creators a flat monthly subscription with clear usage limits, and takes no cut of course sales; the tiers below are hypotheses to test with the first creators, not final prices.

Tier (draft)	For	Includes	Price hypothesis
Starter	Solo teacher testing online sales	Limited storage and delivery, watermark, low-bandwidth mode	Low entry price, monthly

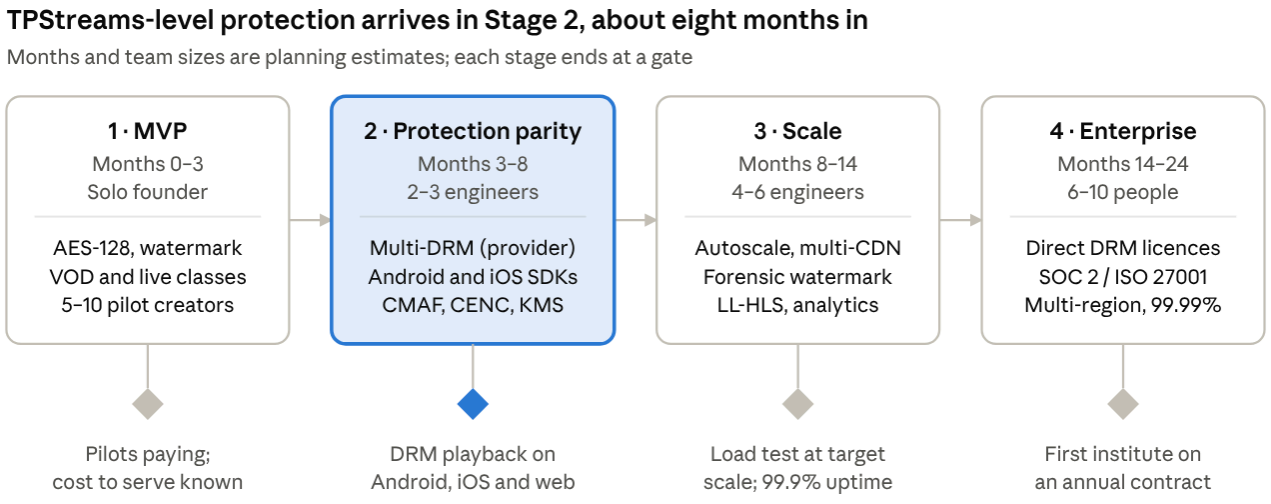
Tier (draft)	For	Includes	Price hypothesis
Pro	Teacher with an active paying batch	More storage and delivery, live classes, analytics	Mid price, monthly
Institute	Coaching centre with several faculty	Multiple staff accounts, higher limits, priority support	Higher price, monthly or annual

Principles

- No revenue share: creators keep their course income, minus the payment gateway's fee.
- Usage above a tier's limit is billed at a published rate, never cut off mid-class.
- Prices are set only after measuring the real cost to serve one student-hour of video and one hour of live class.
- The first 5–10 creators get a free or discounted pilot in exchange for weekly feedback.

Delivery roadmap

Reaching TPStreams' level is a two-year plan in four stages: a solo-built MVP in about three months, then protection parity, scale and enterprise readiness as the team grows.



delivery roadmap · 4 stages over about 24 months

Stage 1 in detail (first 12 weeks, solo). Discovery and creator interviews (week 1), foundation with auth, database, CI and staging (week 2), video core (weeks 3–5), commerce and protection (weeks 6–7), live classes (weeks 8–10), and pilot launch with load and device tests (weeks 11–12).

Why this order. Pilots in Stage 1 prove that creators will pay before money goes into DRM licences and a team. Stage 2 closes the biggest gap with TPStreams, protection, and only then does Stage 3 spend on scale. Each stage's gate decides whether the next stage is funded.

Risks

The two risks that matter most are building more than the timeline allows and creators expecting stronger protection than the MVP provides.

Risk	Impact	Mitigation
Scope grows beyond the planned hours	MVP slips or ships half-finished	Fixed MVP list; every new idea goes to the Next column of the feature table
Creators expect true DRM	Lost deals with larger institutes	Multi-DRM through a licence provider in Stage 2; state MVP protection plainly until then
Live transcoding overloads one server	Classes stutter or drop	Cap concurrent classes; separate live server; load test before pilots
Weak demand from solo teachers	No paying customers	10–15 creator interviews before the build; paid pilots before launch
Rust and media learning curve	Slower early phases	Foundations phase dedicated to Rust, FFmpeg and HLS basics
Security flaw in generated code	Content leaks or free access	Tests first, negative security tests, manual review of sensitive modules